SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1407 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: **Total Marks:50**

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I _____M.Madhu Rushwika(192371066)_________________ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M.Madhu Rushwika*

1) PARSING TABLE CONFLICT IN LL(1)

Analysys

In an LL(1) parser, the parsing table is constructed using the FIRST & FOLLOW sets

=> FIRST determines which terminals can begin strings derived from the production

=> FOLLOW is used when the production can derive the ε (Epsilon)

for each nonterminal & input symbol, the parser expects the only one production.

Example:-

The conflict usually occurs because

=> Two productions have overlapping FIRST set

An ε - production overlap b/w FIRST & FOLLOW sets.

$A \rightarrow aB$

$A \rightarrow ac$

Both have a at their FIRST. so the parser sees that the a, it cannot determine which production to select

=> LL(1) parsing requires one unique production for each table entry. Multiple entries make the grammar as non-deterministic & prevent the parser from making a unique decision using symbols.

→ The grammar should modified by applying
- the technique such as the

⇒ Left factoring
⇒ Elemination of left recursion
⇒ Rewriting the productions:

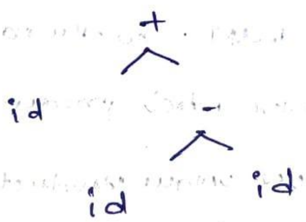
The grammar should be transformed so that the FIRST / FOLLOW conditions required LL(1) are satisfied & every parsing table cell the contains most of the productions.

2) INCORRECT PARSE TREE GENERATION :-

Given Expression,

id + id - id

This contains + & -
If the grammar doesnot specify the associativity, The parser may contains incorrect tree such as these



This Represents id + (id - id)

INVALID EXPRESSION HANDLING :-

Given Input

a+*b

The valid tokens are

a → id
+ → operator
* → operator
b → id

The arrangement is syntactically invalid

A typical Expression grammar might be

$E \rightarrow E+T / T$

$T \rightarrow T*F / F$

$F \rightarrow id$

After reading

a+

The parser expects operand

id or anothe valid Expression instead; it receives

as * so, the parser give syntax error.

A shift reduce parser perfor shift & reduce operation,
when it reaches a state where neithe a valid shift nor
reduction is possible for *, it reports a parsing error

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5)

The parser should provide

→ Clear syntax - error messages

→ Error location

→ Expected token information

→ Error recovery using techniques such as as the

the panic mode recovery.

Syntax Error;

Unexpected '*' after '+'.

Expected identifier '('

The parser should reject the expression because the operator sequence violates the grammar while providing the meaningful error recovery rather than simply stops unexpectedly.

4) AMBIGUITY IN CONDITIONAL STATEMENTS:-

Grammar,

S → if E then S

| if E then S else S

| a

The grammar produces the well known danglings

if E_1 then
    if E_2 then
        a
    else
        a

The else could potentially belongs to

if E_2 or if E_1

Therefore the grammar is ambiguous

=> When constructing the parsing table, the parser can be encounter a situation where it has possible productions

Grammar Modifications

$S \rightarrow M | U$

$M \rightarrow$ if E then M else M

$| a$

$U \rightarrow$ if E then S

$|$ if E then M else U

The grammar should be rewritten to eliminate the ambiguitys or a parser strategy should explicitly apply the nearest if rule so that every else has a unique associations

5) OPERATOR PRECEDENCE CONFLICTS

Grammar

$E \rightarrow E+E$

$\mid E*E$

$\mid id$

Input

id+id*id

The grammar is Ambiguous because it does not specify whether + or * should be evaluated first

(id +id)* id

id+ (id*id)

id +(id* id)

because * > +

precedence Grammar

$E \rightarrow E+T/T$

$T \rightarrow T*F/F$

$F \rightarrow id$

E handles +

T handles *

F handles operands

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precedence Grammar

$E \rightarrow E+T / T$

$T \rightarrow T * F / F$

$F \rightarrow id$

E handles +

T handles *

F handles operands

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and